

How much does the map of science depend on the embedding model?

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Abstract

Changing an embedding model changes the representation of scientific papers. The question is which conclusions change with it. I compare ten models on a fixed corpus of 500,000 OpenAlex records from 26 fields and 2000–2024, using matched panels to examine field centres, internal structure and nearest neighbours, alongside input and representation choices. The models share broad structure but retain only 30.3% of the 25 nearest neighbours on average within field-period groups. Removing abstracts changes neighbours about as much as changing models; removing titles changes less. A later exploratory analysis of all 325 field pairs finds persistent opposite orderings across models in 262 pairs for angular spread and 221 for effective linear dimension. Requiring relative differences above 5% leaves 96 and 177. Alternative measures retain many oppositions, with further dependence on centring and pooling. A field comparison can therefore survive repeated article selection while reversing between models. These findings describe geometry, without establishing thematic correctness. I argue that robustness should be assessed at the level of the claim made from a science map: overall agreement can coexist with disagreement on the comparison that matters.

Keywords: science mapping; document embeddings; robustness; neighbourhood agreement; geometric comparisons.

1 Introduction

The same collection of papers can support different accounts of how a research area is organised. With text embeddings, part of that account comes from the model: it turns each paper into a vector, and relationships between those vectors determine which papers appear close. This is useful for locating related work and comparing fields, as applications to biomedical literature and physics show (González-Márquez et al., 2024; Constantino et al., 2025). I examine what happens when the papers stay the same and that model changes. The question concerns the conclusions drawn from the representation, beyond whether the resulting maps broadly resemble one another.

The concern itself is well established. Gläser et al. (2017) asked how much an inferred thematic structure belongs to scientific knowledge and how much to the method that reconstructs it. In Boyack and Klavans (2020), text, citation and hybrid relations produced substantially different cluster memberships even when aggregate accuracy was similar. That distinction motivates the present comparison. Two maps may agree in a summary score and still lead a reader towards different papers or different comparisons between fields.

I keep a corpus of 500,000 publications fixed: records labelled English in OpenAlex, published in 2000–2024 and assigned to 26 broad Fields. Ten embedding models represent the same papers. The comparisons follow the map from field centres to relationships within fields and then to each article’s neighbours. Separate matched subsets vary the input text and the way a model’s output becomes one document vector. A subsequent geometric extension asks whether changing the encoder reverses a comparison between two fields, even when its direction persists across article selections.

The results make that last distinction consequential. Broad relationships are shared, but individual neighbours and geometric field orderings can change. For some comparisons, repeated article selection supports one direction under one model and the opposite direction under another. I retain both the magnitude of those differences and their dependence on processing, because a persistent direction alone says little about the size of the disagreement. The analysis identifies which relationships survive the changes tested. It leaves their thematic validity for a separate assessment.

2 Background and research questions

2.1 Representation choices in science mapping

Previous comparisons help specify what such an assessment would require. Constantino et al. (2025) evaluated text and graph representations against a hierarchical classification of physics papers, including title and abstract representations. González-Márquez et al. (2024) compared models and output-reduction choices for a large biomedical literature map, using shared field labels among nearby papers as a criterion. In each case, a representation is judged through a particular use and reference. Here I ask whether a conclusion survives a change of representation. Those external evaluations clarify what further evidence would be needed to judge its thematic correctness.

A model name is only the beginning of a representation recipe. The title and abstract contribute different information, input limits determine how much reaches the model, and token outputs may need to be combined into a document vector. Pooling names that last operation: averaging the outputs and selecting a designated position are different choices. Singh and Singh (2022) demonstrated sensitivity to short queries and wording changes in

scientific language models. For the present purpose, leaving these choices unspecified would leave part of the mapping procedure outside the comparison.

Comparing geometry and neighbours together also has direct precedents. Caspari et al. (2024) did so across retrieval datasets; Imel and Hafen (2025) related local density and asymmetry in different document embeddings and disciplines to citation rates. I build on these approaches by following broad, local and geometric conclusions across a fixed panel. The same article identities, explicit text and representation conditions, and controlled candidate pools make the comparison refer to stated changes rather than an unspecified difference between systems.

2.2 Agreement, stability and interpretation

I use agreement and stability for separate questions. Agreement asks what two representations share on the same articles. Stability asks whether a result persists when a specified part of the comparison changes. A field ordering can pass the latter check within every model and still disagree across models. Calling either result thematically valid requires evidence about how the relationships correspond to the research. Held and Velden (2022) make this interpretive issue clear in invasion biology, where mapping methods can combine different relationships between papers. Neither a repeated result nor a consensus supplies that missing interpretation.

The object called *shape* must also be specified. All comparisons here precede projection into two dimensions. Field centres describe broad organisation; the articles within each field describe its internal relationships; neighbours identify which papers remain closest to an article. Angular spread and effective linear dimension add descriptions of vector separation and the distribution of variation across directions. I keep these geometric meanings throughout. Greater spread alone cannot establish more diverse topics, and a dimension value cannot be read as a topic count. Each property answers part of the question about the map.

2.3 Research questions and scope

Three questions organise the analysis:

- RQ1.** Which structural relationships are shared across embedding models, and how does agreement vary between field centres, articles within fields and local neighbours?
- RQ2.** How much do text input and pooling choices change geometry and neighbours relative to changing the embedding model?
- RQ3.** Which geometric comparisons between fields persist, reverse or remain unresolved across models and article selections, and how large are the differences?

The scope is descriptive and limited to the selected articles and models. The third question was added in an exploratory extension after the first comparisons; the methods and supplement retain that chronology. OpenAlex labels organise the work, including their known errors, without providing an independent reference for thematic correctness. I therefore assess the persistence of conclusions within this design. The study has no external criterion with which to name the model that recovers a single correct science map.

3 Data and comparative design

3.1 Corpus and analytical panels

The corpus contains 500,000 distinct OpenAlex work IDs, collected and cleaned by 15 September 2026. OpenAlex provides the metadata and texts (Priem et al., 2022); its core catalogue was restricted to articles, reviews and conference papers published in 2000–2024, labelled English, with a non-empty title, a reconstructible abstract of at least 50 words and a known primary Field. Retraction and paratext flags were excluded. Clear language and content failures were removed, while uncertain cases and possible duplicate versions were retained with flags. Each record’s primary-topic classification was frozen at extraction. These automatically assigned labels use text, citation and journal information (OpenAlex, 2026); they organise the analysis rather than supply an independent standard of correctness.

A general sample of 400,000 eligible records was drawn through seeded random API requests. An additional 100,000 records reinforced the smallest Field-period groups by repeatedly allocating places to the least populated group. Large groups were not reduced to the size of the smallest. The resulting 130 groups cover 26 Fields and five periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Every group contains at least 2,165 records. Global corpus comparisons use the general sample; group comparisons can use both cohorts. Equal-weight averages across groups describe a balanced comparison, not the proportions of world research.

I use separate panels for changes that need different controls (Table 1). The three-input panel contains 52,000 articles balanced across Field-period groups, including the 26,000-article pilot. Another panel of 52,000 compares native inputs with the same literal fragment that fits all tokenizers. Equal-size group comparisons address group size, while matched neighbour searches control the search opportunity. Article identities are matched between models throughout. Supplement S1 records extraction, cleaning, counts and the quality flags left in the corpus.

Table 1: Corpus and analytical panels.

Panel	Articles and representation	Comparison and scope
Original corpus	500,000 IDs: 400,000 base + 100,000 supplement. Ten models; title + abstract.	26 Fields, five periods. Global summaries use the base; the full corpus is not population-proportional.
Three text inputs	52,000 IDs; 400 per Field-period, 2,000 per Field. Title, abstract and both.	Same articles and candidates between models. Includes the 26,000-article pilot.
Common fragment	A different 52,000-ID panel; 400 per Field-period.	Paired native/shared content; model-specific tokenization. Distinct from the three-input panel.
Matched group size	256 articles per centre; 26 Fields, 217 Subfields, or 26 Subfield centres.	Within groups: 26 Fields and the same 183 Subfields at 128, 256 and 512 articles. Different objects from centres.
Paired searches	217 Subfields; 50 fixed queries each (10,850 total). Ten candidate selections.	256 candidates in each scope, including all 50 queries. The Field pool is conditioned on these queries.
Geometry / Field pairs	2,000 per Field; twenty halves of 1,000 and five selections of 2,000.	325 Field pairs and two properties; 26 saved conditions. Alternative metrics have different coverage.

Rows overlap and must not be added. Periods: 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Each model comparison uses the same IDs. Query articles exclude themselves during neighbour search. The 1,300 technical test articles are not an additional scientific panel. Full filters and quotas: Table S1.

3.2 Models and text representations

The ten models provide contrasting scholarly representations, spanning scientific documents, general sentences and word-level biomedical/scientific models (Table 2). I use them as a fixed comparison panel chosen for relevance and training contrasts. They are neither a random sample of encoders nor a ranking of the newest systems. Checkpoints, tokenizer revisions and text formats were fixed before inference. Every model received the same title and abstract through its customary format and native input limit. MiniLM’s native 256-token limit remains primary, with a separate 512-token sensitivity. Native truncation affected 49.58% of MiniLM inputs and 20.16% of MPNet inputs.

For BERT, SciBERT, BioBERT and PubMedBERT, the principal document vector is the mean of last-layer, non-padding token states, including special tokens. These models were not trained specifically for document similarity. Their first-position state (CLS, before the pooler) and final non-padding state (SEP) provide alternative readouts. The mean rule was specified before the primary comparisons and retained after the early sensitivities; it was not selected to maximise agreement. The other six models use their documented representation recipes. Title-only and abstract-only inputs use the same 52,000 IDs as the combined-input comparison. Native limits mean that this tests practical ways of using the models, without isolating architecture from text length.

Table 2: Embedding models and primary representation settings.

Model	Dim.	Tokens	Readout	Reference
SPECTER	768	512	CLS	(Cohan et al., 2020)
SPECTER2	768	512	CLS + adapter	(Singh et al., 2023)
SciNCL	768	512	CLS	(Ostendorff et al., 2022)
SciBERT	768	512	Mean	(Beltagy et al., 2019)
BERT	768	512	Mean	(Devlin et al., 2019)
MPNet	768	384	Mean	(Reimers and Gurevych, 2019)
MiniLM	384	256	Mean	(Reimers and Gurevych, 2019)
PubMedBERT / BiomedBERT	768	512	Mean	(Gu et al., 2022)
BioBERT	768	512	Mean	(Lee et al., 2020)
SimCSE	768	512	CLS, pre-pooler	(Gao et al., 2021)

MPNet and MiniLM use all-mpnet-base-v2 and all-MiniLM-L6-v2, respectively. Limits count tokens including special symbols. Mean: last-layer average over non-padding positions, including special symbols. CLS: first-position state before the pooler; SPECTER2 adds its proximity adapter. For MPNet/MiniLM, the reference describes the Sentence-BERT framework, not the exact check-point training. Readout variants are not additional models. Full revisions, formats and truncation: Table S2.

3.3 Objects and measures

All measurements use the original embedding dimensions, without a two-dimensional projection. Article vectors are first scaled to unit length. Shape agreement is measured primarily by corrected linear centred kernel alignment (CKA), which compares relationships among the same identified articles (Kornblith et al., 2019; Murphy et al., 2024). Higher values indicate greater agreement. The finite-sample correction reduces a known source of inflated alignment; it does not make the complete ratio an exactly unbiased estimator. Procrustes

alignment and correlations between distance ranks provide checks with different sensitivities. Their definitions and the correction formula appear in Supplement S2. These related measures are not independent votes about validity (Williams et al., 2021; Williams, 2024).

A centre is the mean of a group’s unit-length article vectors. This deliberately reduces a collection of articles to one object for the broad comparison. Within-group comparisons retain the article-level variation. To check how much agreement can accompany averaging itself, random groups preserve the observed sizes and publication periods but discard the observed allocation of labels. The random allocation is shared across models.

Neighbour agreement is the fraction of the k closest articles shared by two models. The main choice is $k = 25$, with 10 and 50 as checks. Searches use cosine similarity and exclude the query article itself. The candidate pool is stated for each result: all articles in a Field-period group, an equal-size subset, a matched Field/Subfield pool, or the general corpus. An exact blockwise calculation avoids storing every pairwise distance. Independent full-sort checks verify saved neighbour lists. Neither overlap nor its simple chance-adjusted form removes all effects of candidate composition or size.

Two additional properties describe each Field geometrically. *Angular spread* is the median angle between distinct article vectors. *Effective linear dimension* is the participation ratio of the centred covariance matrix C , $PR = (\text{tr } C)^2 / \text{tr}(C^2)$. It is larger when variation is distributed across more directions. Angular quantiles, angles to the centre, entropy-based effective rank and the number of directions containing 80% of variance serve as alternatives. The latter two use the same covariance spectrum as PR; they do not provide independent thematic validation. Graph-connection measures were also tested, but their synthetic counterexamples prevented an interpretation as general fragmentation (Supplement S6).

3.4 Matched comparisons and stability

The strict Field/Subfield neighbour comparison fixes 50 query articles in each of 217 Subfields. Both searches contain 256 candidates, including those same 50 queries, and identical period counts. Ten selections vary the remaining 206 candidates within the Subfield or its parent Field. Thus, the broader search is conditioned on the queries; it is not an unrestricted draw of 256 Field articles. The design covers 10,850 queries and holds search opportunity constant while varying its scope.

For Field geometry, I compare every unordered pair of the 26 Fields, giving 325 pairs per property. A direction is persistent for a model only when it holds for the primary 2,000 articles per Field, twenty half samples of 1,000 and five further selections of 2,000 from the same corpus. Agreement of all ten requires that common persistent direction in every model. Opposition requires at least two fixed models that persistently support opposite directions. Everything else is unresolved, a category that does not imply equality. Relative magnitude

is $2(B - A)/(A + B)$, with A and B the two Field values. Minima of 0, 1, 5 and 10% show how the classification depends on small differences.

These repetitions test what happens when articles change within the corpus already collected. Their overlapping selections do not produce independent population samples or confidence intervals. The operational screens preserve excessive median shifts and broad selection ranges as alerts, with separate rules and denominators in the supplement. The pair comparisons also share their components: a model belongs to several model pairs, and a Field to several Field pairs. I report their counts descriptively and use no significance test that treats the pairs as independent.

3.5 Sequence of checks

The comparisons were developed in stages, and the order matters for their interpretation. The initial shape and neighbour analysis was followed by the expanded input panel, matched Subfield searches and case atlas. Morphology came later as an exploratory extension; the all-pairs summary followed its initial examples. Each new calculation had rules recorded locally beforehand, but the whole study was not externally preregistered. Some controls, including random-group centres and a fixed-neighbour-fraction comparison, followed earlier results. Supplement S2 keeps that sequence explicit. Supplements S3–S7 contain the time, composition, model-trait, source-error and processing checks, including those that constrain the conclusions.

4 Results

4.1 Shared broad structure and within-field variation

Models agree strongly about the arrangement of group centres. With 256 articles contributing to each centre, mean corrected CKA is 0.939 for the 26 Fields and 0.911 for 217 Subfields (Figure 1A). Restricting the latter to 26 Subfield centres gives 0.902. Random groups with matched sizes and dates also show positive agreement, around 0.63–0.64, but do not recover the observed level. Averaging therefore contributes to the comparison, yet is insufficient to account for the agreement associated with these labelled groups.

The internal comparison gives a flatter result. Mean CKA stays close to 0.63 in Fields and 0.64 in the same 183 Subfields eligible at every size (Figure 1B). Moving from 128 to 512 articles scarcely changes those means. A more specific label consequently does not imply less agreement here. The large difference from the centre panel needs a narrower reading: group centres and individual articles are different objects, so the panels cannot estimate a causal effect of scale.

The alternative measures broadly support the ordering of model-pair agreements. Across the original 130 Field-period groups, the mean rank correlation with corrected CKA is 0.891 for Procrustes and 0.881 for distance ranks. Their agreement is incomplete, and the original selection screen retains 50 alerts among 5,850 model-pair/group comparisons (Table S4). These qualifications concern particular comparisons; an overall mean does not erase them.

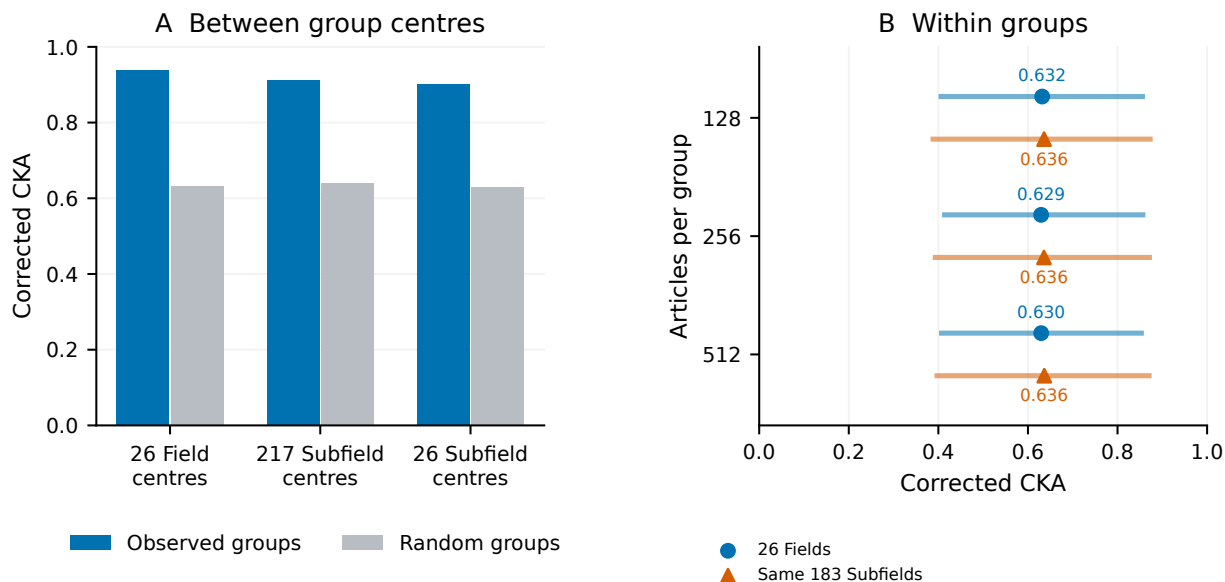


Figure 1: Agreement between group centres and within groups. (A) Mean corrected CKA across the fixed model pairs and saved centre comparisons; each centre contains 256 articles. Random groups preserve sizes and dates. The 26-Subfield condition controls the number of centres. (B) Within-group agreement at three article counts, for 26 Fields and the same 183 eligible Subfields. Points and labels show means; lines span the 2.5th–97.5th percentiles across group–model-pair comparisons (1,170 for Fields; 8,235 for Subfields). These are descriptive spreads, not confidence intervals; model pairs are not independent. All CKA axes span 0–1. The panels compare different objects, so their difference is not an estimated causal effect.

4.2 Neighbour agreement under matched comparisons

Across the full Field-period searches, two models retain an average of 30.3% of the 25 nearest neighbours: about 7.6 articles out of 25. This average gives each group and model pair equal weight. Restricting every group to 2,048 common candidates raises overlap to 31.9%. In the separate search across the 400,000-article general corpus, 13,000 balanced queries retain 17.5%. The last result changes both the scope and the number of available candidates, so it cannot by itself establish weaker agreement across disciplinary boundaries.

The matched Field/Subfield experiment addresses that ambiguity directly (Figure 2). With the same queries, dates and 256-candidate size, Subfield agreement is lower in 127 of 217 cases and higher in 90. The mean difference is small, at -1.35 percentage points. For 50 neighbours, the split becomes 109 lower and 108 higher. A narrower search can preserve more neighbours or fewer, depending on the group and the chosen neighbourhood size. There is no universal loss of agreement from Field to Subfield.

It would also be misleading to treat shape and neighbours as unrelated tests. Across the 45 model-pair means in the initial analysis, their descriptive rank correlation is 0.920. Their joint use has a different justification. CKA describes relationships distributed through the space, whereas neighbour overlap tells us whether a reader would encounter the same articles. Both matter for interpretation even when they move together.

The atlas makes those local differences inspectable. Its Subfield examples are selected by a recorded persistence rule across 27 combinations of size, readout and neighbour count. Article-level examples then show links that persist across models alongside links whose inclusion varies sharply (Supplement S4). Some extreme cases expose source problems: a history book appears under Physics, an infection-related record under Political Science, and an announcement remains in the corpus. Comparison with the original records confirms that these are source or classification issues, rather than mismatched embedding rows. The examples are exploratory and deliberately selected; they estimate neither the prevalence of such errors nor the accuracy of the links.

4.3 Text input and representation choices

An input decision can be as consequential as the choice of encoder. For the same 52,000 articles, changing models gives a mean shape change of 0.370, measured as one minus corrected CKA. Keeping the model but leaving only the title gives 0.366. The corresponding neighbour-replacement fractions are 68.8% and 70.7% (Figure 3). I read these close averages as evidence that the two choices can have comparable consequences within this design. They do not establish equivalence between their effects.

Removing only the title changes less. Abstract-only inputs have a mean shape change of 0.027, yet replace 22.1% of the 25 neighbours. A small change in overall geometry can therefore still alter the papers returned locally. The model-specific results also vary substantially: title-only neighbour replacement ranges from 51.7% for MPNet to 84.2% for SciBERT (Table S5). A single pooled mean would conceal those differences.

That comparison itself depends on pooling. Under the principal mean rule, removing the title has a relatively modest effect, while removing the abstract slightly exceeds model change in mean neighbour replacement. With CLS or SEP for the four word-level BERTs, the latter ordering reverses. These scenarios move four representations at once, including the cross-

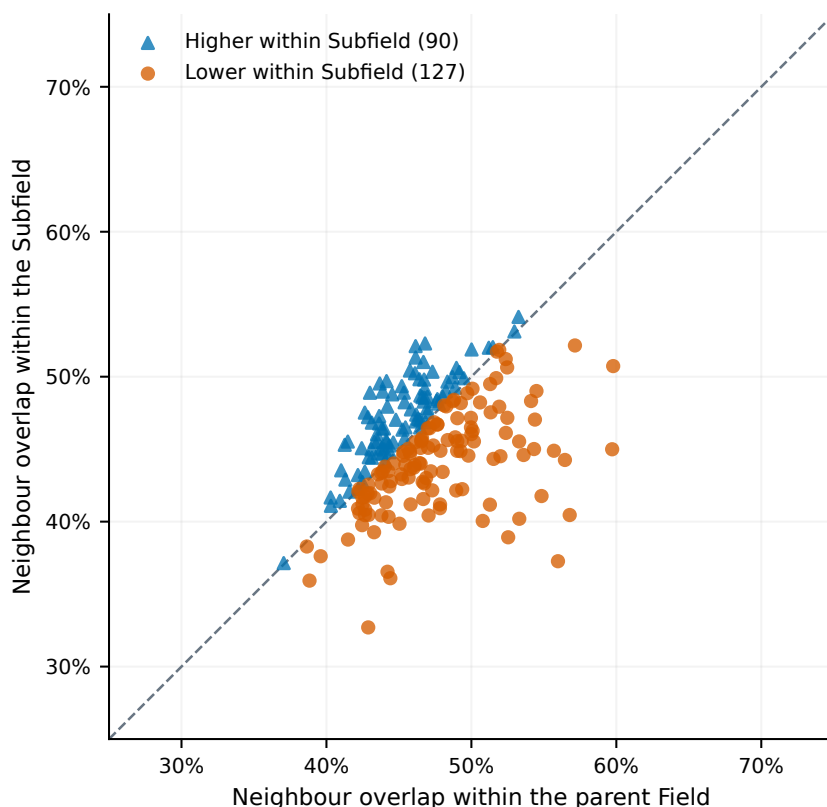


Figure 2: Neighbour agreement in matched Field and Subfield searches. Each point is one of 217 Subfields, averaged over model pairs and ten candidate selections. Both scopes contain the same 50 fixed queries within 256 candidates and matched dates; self is excluded. The broader Field pool is conditioned on these queries. Points above the diagonal have higher overlap in the narrower search; those below have lower overlap. Dispersion describes group differences, not confidence intervals.

model reference. They cannot isolate a pure pooling effect. Selecting the readout that makes the models agree most would change the purpose of the comparison.

The larger input panel reduces sensitivity to article selection. Using 100 comparable selections at each size, the shape-contrast screen drops from 72 alerts among 520 model–Field–input contrasts at 26,000 articles to five at 52,000. The earlier 20-selection check gives 31 and four, respectively (Figure S1). Both records are retained. Passing this screen does not mean that a contrast differs from zero, nor does it certify all neighbour results or readout variants. The common-fragment and MiniLM-length checks further distinguish unequal text exposure from a coding error: the original MiniLM limit was its documented setting, not an implementation fault.

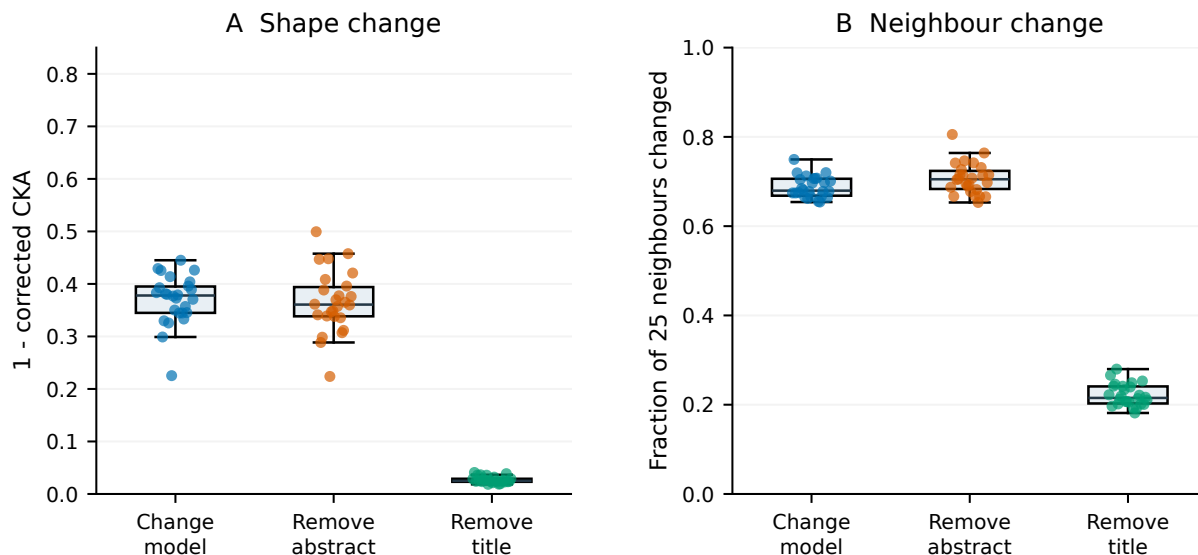


Figure 3: Changes associated with model and text input. The same 52,000 IDs provide 2,000 candidates per Field. The reference input is title plus abstract: “remove abstract” leaves the title, and “remove title” leaves the abstract. (A) One minus corrected CKA. (B) Fraction of the 25 neighbours replaced. Each point is a Field mean over the fixed model panel. Boxes show medians and interquartile ranges, with 1.5-IQR whiskers; all Field points are displayed. These are not confidence intervals. Native text limits are retained.

4.4 Persistence and reversal of geometric field comparisons

The geometry extension asks what happens to a specific statement about two Fields when its representation changes. Before comparing models, article selection provides an internal check. Angular spread retains no alerts across the 260 model–Field cases in either the half-sample or equal-size screen. Effective linear dimension retains four half-sample alerts, though none occurs in the five equal-size selections (Table S12). I keep those four cases visible; the second check does not cancel the first.

Stability within a model does not ensure agreement between models. Among all 325 Field pairs, angular spread has a persistent common direction in all ten models for 42 pairs, persistent opposition for 262 and an unresolved outcome for 21. Effective linear dimension has corresponding counts of 54, 221 and 50 (Figure 4). Opposition means that at least two particular models repeatedly support opposite directions across all 26 saved selection conditions. It does not require every model to disagree, and unresolved pairs can contain partial agreement.

Many angular differences are small. Requiring each opposing direction to exceed a symmetric relative difference of 5% reduces angular opposition to 96 pairs; at 10%, 19 remain. Effective

linear dimension retains 177 and 126 oppositions at those cutoffs. Thus, the direction-only totals should not be read as counts of equally large disagreements. The minimum is an explicit sensitivity choice, not a universal definition of a meaningful scientific difference. Figure 4 retains all 325 pairs in each row so that the growing unresolved category remains visible.

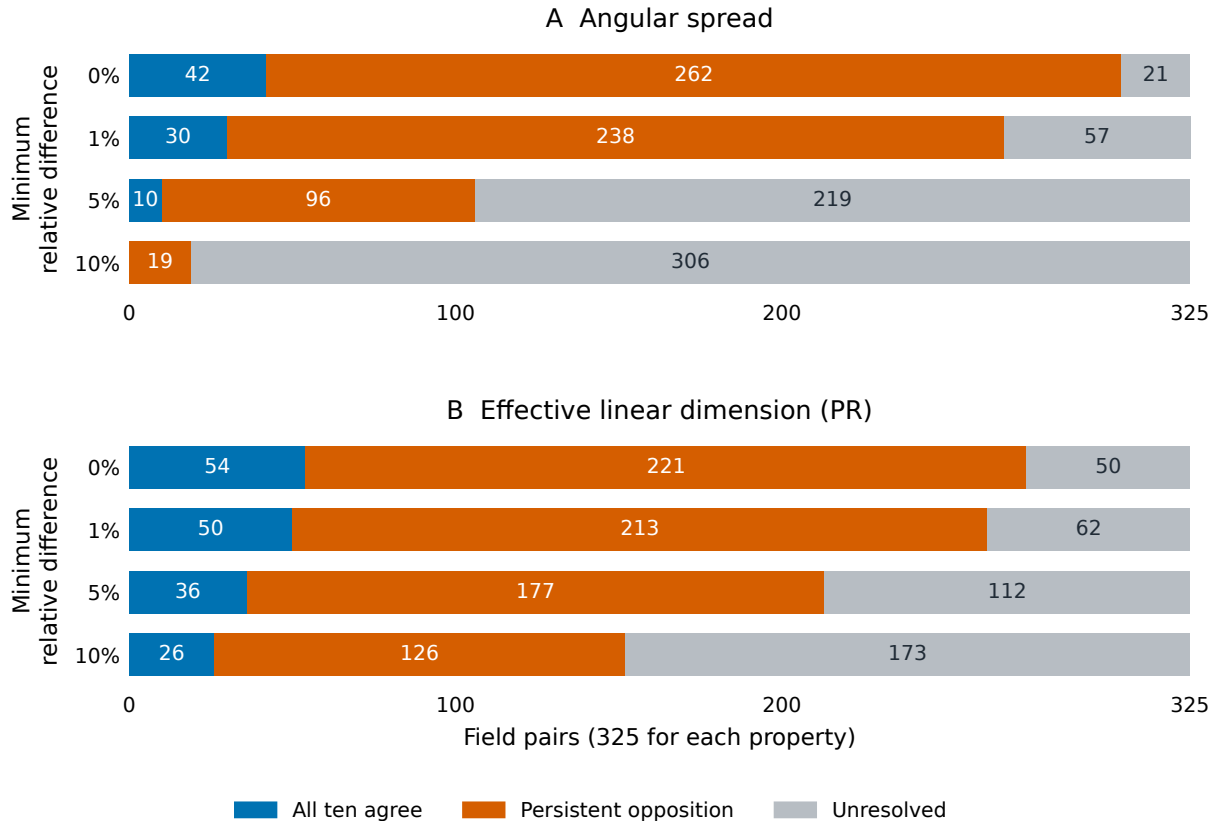


Figure 4: Persistence, opposition and magnitude of Field comparisons. Each row contains all 325 Field pairs for one property and ten models. A model direction must persist across the primary selection, twenty half samples and five additional equal-size selections from the same corpus. All-ten agreement requires a common persistent direction; opposition requires at least two fixed models with persistent opposite directions. All remaining pairs are unresolved. Magnitude uses $2(B - A)/(A + B)$. The four minima are operational cutoffs, not significance or accuracy thresholds. The 26 conditions overlap; point controls are reported separately.

Alternative measures preserve much of the pattern. Requiring the same opposing models to keep their directions across the angular alternatives retains 225 of the original 262 oppositions. The spectral alternatives retain 196 of 221. Their coverage differs: angular alternatives have all 26 selection conditions, whereas entropy-based rank and the 80%-variance count are available for three. These alternatives also share much of their mathematical construction. Their agreement supports a geometric description without turning it into a measure of topic diversity.

The result is not confined to the four word-level BERTs. Restricting the panel to SPECTER, SPECTER2, SciNCL, MPNet, MiniLM and SimCSE still yields 231 angular and 160 effective-dimension oppositions at zero minimum. A smaller panel has fewer opportunities to contain opposing models, so the change in counts does not isolate an effect of training objective. All model omissions and the complete pair classifications accompany Tables S14–S15.

Processing choices set a further boundary. After subtracting the global centre and renormalising, the same original opposing model pair and directions survive for 145 of 262 angular oppositions and 218 of 221 effective-dimension oppositions in the matched point check. Angular ordering is therefore more dependent on that choice. Pooling changes can also reverse comparisons (Table S16). These are point sensitivities with matched references, not another set of 26-repeat confirmations. The retained conclusions concern explicitly defined representations, rather than properties expected to survive any transformation.

Connection did not meet the interpretive requirement. Synthetic elongated clouds and isolated extreme points could produce signals resembling fragmentation, while sample size and neighbour count substantially altered the measurements. The equal-size screen retained 51 connection alerts among 260 cases. I therefore report the connection results and their failed construct checks in Supplement S6, without using them to infer that a Field contains more separate thematic groups.

5 Discussion

5.1 From map agreement to the persistence of a claim

The most useful result is the coexistence of shared structure and persistent disagreement on a particular claim. Centre agreement exceeds the random-group reference. Yet models that share that broad organisation can repeatedly order a pair of Fields in opposite ways as the articles change. This is why I would attach a robustness assessment to the comparison being interpreted. A stable neighbour set, a relationship between groups and an ordering on a geometric property each require their own check; an overall score cannot settle all of them.

That reading builds on an established concern in science mapping (Gläser et al., 2017; Boyack and Klavans, 2020). Constantino et al. (2025) evaluate representations against a specified external classification. Caspari et al. (2024) compare the shared geometry and retrieval relationships of representations. Here I connect these levels to the persistence and size of concrete Field comparisons under matched articles. The contribution rests on that connection and its controls. The measures of agreement and the concern with representation choice both have clear precedents.

The geometric measures are helpful within that deliberately limited interpretation. An angle describes separation; PR describes how variance occupies linear directions. Calling either

interdisciplinarity or semantic diversity would add a claim about content that these calculations have not tested. Work on embedding isotropy raises a related problem of matching a geometric quantity to its interpretation (Rudman et al., 2022). The connection checks show the practical cost: repeatable differences between clouds were insufficient grounds for the proposed fragmentation reading.

5.2 Implications for comparative science mapping

For a science-mapping application, I would begin with the comparison that the reader is expected to interpret. A retrieval claim needs a specified candidate pool and neighbour count. A geometric Field comparison needs an explicit representation rule, article balance and minimum difference. These choices should guide the robustness check. The input experiment also gives a concrete reporting lesson: the model name should be accompanied by the title/abstract choice, text limit and pooling rule, because each helps determine the representation being interpreted.

The supplementary analyses resist an easy account of why models disagree. Medicine has lower shape agreement than Energy, yet is not consistently lowest for neighbours. The Medicine–Energy difference remains after balancing Subfields and after removing BioBERT and PubMedBERT (Supplement S5). Those checks narrow possible explanations without establishing a unique cause. Model families are also readout-dependent. Since architecture, training source and objective vary together across these encoders, the observed similarities cannot assign a separate causal role to any one of them.

Time makes the dependence on comparison design especially clear. Mean shape agreement increases slightly between the earliest and latest periods. Mean neighbour agreement falls by 0.87 percentage points with all candidates and rises by 0.91 points with 2,048 candidates in each group. The search opportunity changes the sign. I would therefore avoid reading this as evidence that science has historically converged. The models use current parameters to represent old and new publications; the calculation does not reconstruct the knowledge available when those papers appeared.

5.3 Limits and next validation

The claims stop at the corpus as well as the comparison. These are selected OpenAlex records with usable abstracts and predominantly English text. Reinforcement, balanced panels and equal group weights support comparisons between Fields, without estimating the composition of world research. The atlas identifies classification and content errors that survive cleaning. The checks address named failure modes rather than certify every record. Saved selections and checkpoints make the work inspectable, but repeating selections cannot correct what the corpus fails to cover.

The ten encoders provide a bounded set of alternatives. Several share architectures, training sources and objectives, and several are not specialised similarity models. Restricting the analysis to six similarity-oriented encoders or changing readouts addresses some concerns without producing a representative sample of models. Persistent opposition establishes a dependence within this panel. Deciding which direction better describes the underlying research would require evidence beyond that opposition.

The sequence of the work is another limit I retain. The geometry extension and all-pairs summary followed earlier findings; complete records do not change their exploratory status. For the next step, I would choose one consequential comparison and assess it against independently judged article relationships or a discipline-specific question. This would address the connection between geometry and subject matter that remains open. Adding encoders alone would extend the agreement comparison without supplying that connection.

6 Conclusion

Broad agreement between models can coexist with different neighbours and opposite geometric orderings of Fields. Input choice can be as consequential as changing the model, and repeated article selection can preserve a result that reverses between encoders. I therefore favour testing the claim drawn from the map alongside overall similarity. Angular spread and effective linear dimension make that possible for specific geometric comparisons, provided their definitions, magnitudes and processing sensitivities stay explicit. The remaining step is substantive: interpreting those model-produced geometries as properties of scientific knowledge requires further validation.

Declarations

Author contributions. Alejandro Treny Ortega conceived the project and directed the research and manuscript preparation as an independent researcher. Computational and editorial assistance is disclosed below.

Competing interests. The author declares no competing interests.

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Data and code availability. The project repository is <https://github.com/aleetreney/Shape-of-Science-Robustness>. This draft is accompanied by the tables used for its figures and the editable manuscript sources. The frozen corpus, vectors and complete local execution records have not been deposited in a persistent public archive. Public code alone does not provide full reproduction of the present analysis. The access inventory and

remaining release requirements are stated in Supplement S8; no data deposit or persistent identifier is claimed here.

Use of AI-assisted tools. OpenAI Codex assisted with literature searches, software development and analysis, visualisation, and manuscript drafting and editing. The author is responsible for the content.

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